Medical Laboratory Techniques

Level III

TVT Curriculum Based on Version 4

Occupational Standard

Prepared By Ministry of Labor and Skills

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Addis Ababa, Ethiopia

Preface

The reformed TVET-System is an outcome-based system. It utilizes the needs of the labor market and occupational requirements from the world of work as the benchmark and standard for TVET delivery. The requirements from the world of work are analyzed and documented taking into account international benchmarking as occupational standards (OS).

In the reformed TVET-System, curricula and curriculum development play an important role with regard to quality driven TVET-Delivery. Curricula help to facilitate the learning process in a way, that trainees acquire the set of occupational competences (skills, knowledge and attitude) required at the working place and defined in the occupational standards (OS). Responsibility for Curriculum Development will be given to the Regional TVET-Authorities and TVET-Providers.

This curriculum has been developed by a group of experts from different Regional TVET-Authorities based on the occupational standard for Medical Laboratory Techniques Level IV. It has the character of a model curriculum and is an example on how to transform the occupational requirements as defined in the respective occupational standard into an adequate curriculum.

The curriculum development process has been actively supported and facilitated by the Ministry of Health and Ministry of labor and skill.

TVET-Program Design

1.1. TVET-Program Title: Medical Laboratory Techniques Level IV

1.2. TVET-Program Description

The Program is designed to develop the necessary knowledge, skills and attitude of the learners to the standard required by the occupation. The contents of this program are in line with the occupational standard. Learners who successfully completed the Program will be qualified to work as a Medical Laboratory Technician with competencies elaborated in the respective OS. Graduates of the program will have the required qualification to work in the health sector in the field of Medical Laboratory.

The prime objective of this training program is to equip the learners with the identified competences specified in the OS. Graduates are therefore expected to Perform Hematological Tests, Perform Microbiological, Use Info-technology Devices in the Workplace, Perform Serological Tests, Perform Clinical Chemistry Tests, Perform Immuno-Haematological Tests, Implement Laboratory Quality Assurance, and Prepare Histopathological Samples for Examination, Manage Community Health Service, Prevent and Eliminate MUDA in accordance with the performance criteria described in the OS.

1.3. TVET-Program Learning Outcomes

The expected outputs of this program are the acquisition and implementation of the following units of competence:

HLT MLT4 01 1121 Use Info-technology Devices in the Workplace
HLT MLT4 02 1121 Perform Microbiological
HLT MLT4 03 1121 Perform Hematological Tests
HLT MLT4 04 1121 Perform Serological Tests
HLT MLT4 06 1121 Perform Clinical Chemistry Tests
HLT MLT4 05 1121 Perform Immuno-Haematological Tests
HLT MLT4 07 1121 Prepare Histopathological Samples for Examination
HLT MLT4 08 1121 Implement Laboratory Quality Assurance
HLT MLT4 09 1121 Manage Community Health Service

HLT MLT4 10 1121 Prevent and Eliminate MUDA

1.4. Duration of the TVET-Program

The Program will have duration of 1686 hours including the on-the-job practice or cooperative training time and Civic Education et al.

s.no	Unit competency	On school training		Cooperative training	Total hours	Remarks
		Theory	Practical			
4.	Use Info-technology Devices in the Workplace	40	30	-	70	
5.	Perform Microbiological	200	70	80	350	
6.	Perform Hematological Tests	120	70	60	250	
7.	Perform Serological Tests	100	50	50	200	
8.	Perform Clinical Chemistry Tests	120	70	60	250	
9.	Perform Immuno-Haematological Tests	90	40	50	180	
10.	Prepare Histopathological Samples for Examination	80	40	20	140	
11.	Implement Laboratory Quality Assurance	50	20	20	90	
12.	Manage Community Health Service	50	20	30	100	
13.	Prevent and Eliminate MUDA	24	16	16	56	
14.	Total hours	874	426	386	**1686**	

1.5. Qualification Level and Certification

Based on the descriptors elaborated on the Ethiopian National TVET Qualification Framework (NTQF) the qualification of this specific TVET Program is “Level IV”.

The learner will not be awarded any certificate before completion of all the modules that are designed for the exit in level IV.

1.6. Target Groups

Any citizen who meets the entry requirements set by the concerned organization for the academic year and capable of participating in the learning activities is entitled to take part in the Program.

1.7 Entry Requirements

The prospective participants of this program are required to possess the requirements or directive of the ministry of labor and skill.

1.8 Mode of Delivery

This TVET-Program is characterized as a formal Program on middle level technical skills. The mode of delivery is co-operative training. The TVET-institution and identified companies have forged an agreement to co-operate with regard to implementation of this program. The time spent by the trainees in the industry will give them enough exposure to the actual world of work and enable them to get hands-on experience.

The co-operative approach will be supported with school-based lecture-discussion, simulation and actual practice. These modalities will be utilized before the trainees are exposed to the industry environment.

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1.9. TVET Program Structure

Unit of Competence	Module Code & Title		Learning Outcomes	Duration (In Hours)
Use Info-technology Devices in the Workplace	HLT MLT4 M01 02222	Using Info-technology Devices in the Workplace	• Info Techniques systems • computer-based equipment and systems • Store and present files/data • Work place procedure for management and security of data. • Shut down computer	70
Perform Microbiological Tests	HLT MLT4 M02 1224	Microbiological Tests	• Concept of microbiology • Process sample • Microbiology safe work and aseptic technique applications. • Prepare and perform direct examination • Perform examination of stained slides • Prepare culture media • Sterilize media • Pour, label and store media • Perform quality control checks • Maintain records of laboratory	360

		work	
Perform Hematological Tests	HLT MLT4 M03 1224 Haematological Tests	• Basics of Immunohematology • Process immunohematological samples • Types of Immuno Haematological tests • safe work environment • Blood Donation and Collection • Hemolytic Disease • Laboratory records	180

Perform Serological Tests	HLT MLT4 M04 1224	Serological Tests	• Concept of immunology and serology • Process serological sample • Tests request • Laboratory records • Safe work environment	200
Perform Clinical Chemistry Tests	HLT MLT4 M05 1224	Clinical chemistry Tests	• *Basics of clinical chemistry* • Test requirements and processing • Equipment and instrument management • Classifications of clinical chemistry tests • Data processing and interpretation • Data quality in clinical chemistry • Maintaining a safe working environment and hazard identification	250

Perform Immuno Haematological Tests	HLT MLT4 M06 1224	Immuno Haematological Tests	• Identify concept of Immunohematology • Process samples and associated request forms • Perform tests • Maintain a safe environment • Maintain laboratory records	180
Prepare Histopathological Samples for Examination	HLT MLT4 M07 1224	Preparing Histopathological Samples for Examination	• Introduction to histology • Equipment and materials setup • Tissue processing • Staining sections • safe working environment	140
Implement Laboratory Quality Assurance	HLT MLT4 M081224	Quality assurance	• Identifying the Concept quality assurance • Preparing document and record • Implementing quality Pre-analytic process • Implementing quality analytic process	90

			• Implementing quality post-analytic process • Conducting Process improvement activities	
Manage Community Health Service	<u>HLT MLT4 M09 1224</u>	Managing Community Health Service	• Follow organizational guidelines, understand health policy and service delivery system • Plan, manage, monitor and evaluate health system • Lead and build individual's and team's capacity	100
Prevent and Eliminate MUDA	<u>HLT MLT4 M10 1224</u>	Preventing and Eliminating MUDA	• Prepare for work. • Identify MUDA and problem • Analyze causes of a problem • Eliminate MUDA and Assess effectiveness of the solution. • Prevent occurrence of wastes and sustain operation.	56

*The time duration (Hours) indicated for the module should include all activities in and out of the TVET institution.

1.10 Institutional Assessment

Two types of evaluation will be used in determining the extent to which learning outcomes are achieved. The specific learning outcomes are stated in the modules. In assessing them, verifiable and observable indicators and standards shall be used.

The ***formative assessment*** is incorporated in the learning modules and form part of the learning process. Formative evaluation provides the trainee with feedback regarding success or failure in attaining learning outcomes. It identifies the specific learning errors that need to be corrected, and provides reinforcement for successful performance as well. For the teacher, formative evaluation provides information for making instruction and remedial work more effective.

Summative Evaluation the other form of evaluation is given when all the modules in the program have been accomplished. It determines the extent to which competence have been achieved. And, the result of this assessment decision shall be expressed in the term ‘competent or not yet competent’.

Techniques or tools for obtaining information about trainees’ achievement include oral or written test, demonstration and on-site observation.

1.11 TVET Teachers Profile

The teachers conducting this particular TVET Program are A Level and have satisfactory practical experiences or equivalent qualifications.

2. Learning Module Design

Module Code and Title	HLT MLT4 M01 122422: Using Info-Techniques Devices in the Workplace
Nominal Duration:	70 hours

Module Description: This module covers the knowledge, skills and attitude required to use devices in the workplace including identifying info Techniques equipment and systems ; setting up and shutting down equipment for use ; and inputting, retrieving and presenting files/data in accordance with work requirements.

LEARNING OUTCOMES

At the end of the module the learner will be able to:

LO1. Identify info Techniques systems

LO2. Access and operate computer-based equipment and systems

LO3. Store and present files/data

LO4. Implement work place procedure for management and security of data.

LO5. Shut down computer

Module Contents:

LO1. Identify info Techniques systems

1.1. Identifying types of info Techniques equipment

1.2. Identifying functions of equipment, component parts and accessories

1.3. Interpreting applications for workplace activities of the different info Techniques equipment and systems

1.4. Identifying routine faults in operating systems, software applications and operator errors

1.5. Identifying sources of information on rectifying/reporting faults with operating equipment, systems and application

LO2. Access and operate computer-based equipment and systems

2.1. Adjusting work environments

2.2. Accessing and checking systems for viruses

2.3. Setting up equipment for work requirements

2.4. Using operating manuals and/or help screens for info Techniques equipment and software

2.5.Selecting and Accessing software packages and accessories

2.6. Identifying required file and/or data to be accessed

2.7. Filling files/data

2.8. Following Shut-down procedures for files, applications and equipment

LO3. Store and present files/data

3.1. Entering data using appropriate equipment

3.2. Confirming accurate input

3.3. Accessing files

3.4. Manipulating data and checking for accuracy

3.5. Accessing saved files through relevant directories

3.6. Storing information and disk(s)

3.7. Presenting and saving information

LO4. Implement work place procedure for management and security of data.

4.1. Following security procedures

4.2. Following precautions against the loss or corruption of data

LO5.Shut down computer

5.1. Closing all open applications

5.2. Shutting down computer

Learning Methods:

- Interactive lecture
- Group discussion
- Demonstration
- Simulation
- Group assignment
- Individual assignment

Assessment Methods:

- Continuous assessments (quiz, assignment, tests etc.).
- Oral questioning
- Skill check out
- Final written exam

ASSESSMENT CRITERIA:

ASSESSMENT CRITERIA:

LO1. Identify info Techniques systems

- Types of info Techniques equipment used in the work area are identified
- Functions of equipment, component parts and accessories are identified
- Applications for workplace activities of the different info Techniques equipment and systems are interpreted
- Routine faults in operating systems, software applications and operator errors are identified
- Sources of information on rectifying/reporting faults with operating equipment, systems and application are identified

LO2. Access and operate computer-based equipment and systems

- Work environments and equipment are adjusted to meet ergonomic requirements and workplace policy and procedures
- Systems are accessed and checked where required for viruses
- Equipment are set up for work requirements in accordance with workplace procedures and manufacturers guidelines
- Operating manuals and/or help screens for info Techniques equipment and software are used to inform work practices
- Software packages and accessories for required application are selected and accessed
- Required file and/or data to be accessed is identified
- Files/data are filed according to workplace
- Shut-down procedures for files, applications and equipment are followed

LO3. Store and present files/data

- Data is entered using appropriate equipment, keyboard/mouse, bar code reader, touch screen or other system
- Accurate input is confirmed
- Files are accessed in accordance with workplace procedures
- Data is manipulated to suit work requirements and checked for accuracy
- Saved files are accessed through relevant directories
- Information and disk(s) are stored where appropriate
- Information is presented using computerized projection saved where necessary.
  - Health techniques are used to enhance efficient utilization of resources and avoid duplication of efforts
  - New and/or upgraded equipments are identified, classified and used where appropriate, for the benefit of customers as well as the health care system.

LO 4.Apply the functions of technology

LO4. Implement work place procedure for management and security of data.

- Security procedures are followed in accordance with workplace procedures
- Precautions against the loss or corruption of data are followed in accordance with workplace procedures

LO5. Shut down computer

- All open applications are closed.
- Computer is shut down according to user procedures.

Module Code and Title	HLT MLT4 M02 0324 **Performing Microbiological tests**
Nominal Duration:	**360** Hours

MODULE DESCRIPTION: This module describe the knowledge, skills and attitude required to identify micro-organisms such as bacteria, fungi, viruses and protozoan's using staining techniques and direct examination procedures, and to prepare culture media for culture, isolation and identification of micro-organisms in order to investigate the physiology and pathology of human. This module of competency does not cover the ability to perform Microbiological tests that required advanced knowledge, skills, technology.

TRAINEE'S OUTCOMES

At the end of the module the learner will be able to:

- Explain Concepts of microbiology
- Perform Accurately process sample
- Apply Microbiology safe work and aseptic technique applications
- Prepare and perform direct examination
- Perform stained samples Examination
- Perform Preparation of Culture media
- Apply Technique of media Sterilization
- Discuss Culture media Pouring, labeling and storing
- Implement Quality control in microbiology
- Maintain Laboratory Records and documents

MODULE CONTENTS:

Unit one : Concepts of microbiology

1.1 Fundamental Microbiological Concept

1.2 Microbial Classification and Consequences

1.3 Methodologies for Microbiology Test

1.4 Microscope set up and use

Unite Two :Process sample

2.1 Acceptance Procedures for Samples and Request Form

2.2 Handling Non-Compliant Samples and Request Forms

2.3Recording Samples and Establishing Chain of Custody

2.4 Distribution and Dispatch of Samples

2.5 Proper Storage Protocols for Samples

Unit one 3. Microbiology safe work and aseptic technique applications

3.1 Setting Up Work Area and Essential Equipment

3.2 Implementing Protective Apparel Protocols

3.3 Applying Proper Disinfection Techniques

3.4 Identifying Emergency Equipment Locations

3.5 Adhering to Standard Precautions

3.6 Mitigating Aerosol Release with Biological Safety Cabinets

3.7Cleaning Procedures and Spill Reporting Protocols.

3.8 Practicing Hand Hygiene Pre and Post-Lab Activities

3.9 Safely Disposing of Biohazardous Materials and Laboratory Waste

Unit four. Prepare and perform direct examination

4.1 Preparation of Liquid Films (Wet Mount) for Specimens

4.2 Sample Preparation Techniques for Microscopy

4.3 Conducting Wet Film Examinations

4.4 Documentation and Recording of Examination Results

4.5 Verification of Examination Results

4.6 Proper Storage Procedures for Unused Samples or Sample Components

Unit five . Perform examination of stained slides

5.1 Choosing Staining Techniques

5.2 Preparation of Films/Smears for Staining

5.3 Application of Staining Techniques to Prepared Films

5.4 Conducting Examination of Stained Smears

5.5 Documentation and Recording of Examination Results

5.6 Verification of Examination Results

5.7 .Proper Storage Procedures for Unused Samples or Sample Components for Future Reference and Retesting

Unit six : Prepare culture media

6.1 Preparation of Media-Solvent Mixture

6.2 Media Labeling Procedures

6.3 Utilizing Large Vessels for Mixing and Heating Media

6.4 Dispensing Media into Vessels

Unite Seven: Sterilize media

7.1 Preparing the Sterilizer Unit

7.2 Placement of Sterilization Indicators

7.3 Execution of Sterilization Cycle

7.4 Cooling Media Post-Sterilization

Unit eight: Pour, label and store media

8.1 Incorporating Heat-Labile Constituents

8.2 Ensuring Homogeneous Mixing of Additives and Media

8.3 Dispensing Prepared Media

8.4 Labeling Media Containers

8.5 Proper Storage of Media

8.6 Dating Batch Media

8.7 Incubating Control Plates

Unit nine.: Perform quality control checks

9.1 Media Inspection Protocol

9.2 Assessing Usability of Selective Media

9.3 Verification of Stored Stocks

Unit ten : Maintain records of laboratory work

10.1 Accurate Data Entry and Record Keeping

10.2 Consistent Maintenance of Instrument Logs

10.3 Ensuring Security and Confidentiality of Records

Learning Methods:

- Interactive lecture
- Group discussion
- Role playing
- Case study
- Practice with exercise
- Problem based learning

Assessment Methods:
- Continuous assessments (quiz, assignment, tests etc.).   - Oral questioning (Interview)   - Presentation   - Practical assessment by direct observation of tasks using prepared checklist   - Final written exam
Assessment Criteria

Unit one: Concepts of microbiology

- Explain Fundamental Microbiological Concept
- Identify Microbial Classification and Consequences
- 1.3Discuss Methodologies for Microbiology Test
- 1.4 Perform Microscope set up and use

Unite Two: Process sample

2.1 Accept Procedures for Samples and Request Form

2.2 Handle Non-Compliant Samples and Request Forms

2.3 Record Samples and Establishing Chain of Custody

2.4 Distribute and Dispatch of Samples

2.5 Proper Storage Protocols for Samples

Unit three: . Microbiology safe work and aseptic technique applications

3.1 Set up Work Area and Essential Equipment

3.2 Implement Protective Apparel Protocols

3.3 Apply Proper Disinfection Techniques

3.4 Identify Emergency Equipment Locations

3.5 Adhere to Standard Precautions

3.6 Mitigate Aerosol Release with Biological Safety Cabinets

3.7 Clean Procedures and Spill Reporting Protocols.

3.8 Practice Hand Hygiene Pre and Post-Lab Activities

3.9 Safely Dispose of Bio hazardous Materials and Laboratory Waste

Unit four: Prepare and perform direct examination

4.1 Prepare of Liquid Films (Wet Mount) for Specimens

4.2 Sample Prepare Techniques for Microscopy

4.3 Conduct Wet Film Examine

4.4 Document and Record of Examination Results

4.5 Verify of Examine Results

4.6 Proper Storage Procedures for Unused Samples or Sample Components

Unit five: Perform examination of stained slides

5.1 Choose Staining Techniques

5.2 Prepare of Films/Smears for Staining

5.3 Application of Staining Techniques to Prepared Films

5.4 Conduct Examination of Stained Smears

5.5 Documentation and Record of Examination Results

5.6 Verify of Examination Results

5.7 .Proper Storage Procedures for Unused Samples or Sample Components for Future Reference and Retesting

Unit six: Prepare culture media

Module Code and Title :	HLT MLT4 M03 1224Hematological Tests
NOMINAL DURATION :	**250 Hours**

MODULE DESCRIPTION: This module covers knowledge, skills and attitude required toidentify concepts of physiology and anatomy of human hematopoietic organs, function, activity and interactions of cellular and plasma components of blood, and principle of testing methodology, and prepares samples for basic hematological tests.

LEARNING OUTCOMES

- At the end of the module the learner will be able to:
- Describe the basic hematological concepts
- Identify samples collection procedures and Management
- Perform basic hematology tests
- Define Automation
- Maintain safe working environment
- Maintain laboratory records and documents

MODULE CONTENTS:

Unit one: Basic Concept of Hematology

1.1. Over view of Physiology and Anatomy

1.2. Composition of blood

1.3. Function of blood

1.4. Formation of blood

1.5. Regulation of blood formation

1.6. Formation of RBS

1.7. Formation of WBC

1.8. Formation of Platelets

1.9. Maturation Characteristics of blood cells

Unit two:Collection of samples

- **Phlebotomy**
- Structure of Blood Vessels
- Common Veins Used for Phlebotomy
- Capillary blood collection

Unit three:Basic hematology tests

3.1. Staining blood film

3.1.1Romanowsky stains

3.1.2 Panoptic stain

3.2 Principle of Staining

3.3 Preparation of blood film

3.3.1 Thine blood film

3.3.2 Thick blood film

3.4 White blood cell count

3.4.1 Differential count

3.5. performing Complete Blood Count (CBC) (Manual and Automated method)
3.6. Conducting Hemoglobin (Hg)test
3.7. Carrying out Hematocrit (HCT) determination
3.8. Determine Erythrocyte Sedimentation Rate (ESR)
3.9. Red Blood Cell (RBC) indices
3.9.1 MCV
3.9.2 MCH
3.9.3 MCHC
3.10. Interpreting and reporting test results
3.11 Recording results.
3.12. Verifying results
3.13. Performing communication of test results
3.14. Retaining tested sample or sample components

Unit Four: Automation

4.1 Type of automation
4.2 Principles of automation
4.3 General histogram characteristics of the automation

Unit Five:Safe work environment

5.1 Cleaning up Spills
5.2 Minimizing generation of wastes
5.3 Ensuring safe disposal of bio-hazardous materials and other laboratory wastes

Unit Six:Laboratory records

6.1Entries on report forms and computer systems
6.2 Update instrument maintenance logs
6.3Security and confidentiality

LEARNING METHODS:

- Lecture
- Demonstration
- Simulation
- Exercise
- Individual assignment

ASSESSMENT METHODS:

- Interview
- Written test
- Demonstration/Observation

ASSESSMENT CRITERIA:

Unite One: Basic Concept of Hematology

- Discuss Physiology and Anatomy related to hematology
- Identify Composition of blood

- Explain Function of blood
- Discus Formation of blood
- Explain Regulation of blood formation
- Discus Formation of RBS
- Discus Formation of WBC
- Discus Formation of Platelets
- Explain Maturation Characteristics of blood cells

Unite two:Collection of samples

- Determine site for **Phlebotomy**
- Explain Structure of Blood Vessels
- Identify Common Veins Used for Phlebotomy
- Perform Capillary blood collection

Unite Three: Perform basic hematology tests

- Perform Staining blood film
- Explain Principle of staining
- Discus Romanowsky stains
- Discus Panoptic staining
- Identify blood film
- Perform Thine blood film
- Perform Thick blood film
- Complete White blood cell count
- Complete Differential count
- Perform Hemoglobin (Hg) test
- Perform Hematocrit (HCT) determination
- Perform Erythrocyte Sedimentation Rate (ESR)
- Calculate red Blood Cell (RBC) indices
    - Calculate MCV
    - Calculate MCH
    - Calculate MCHC

- Interpret tests results
- Complete Recording of test results
- Conduct Result verification
- Apply Communication of test results
- Conduct Preservation of sample and sample components

UniteFour:-Automation

- Describe the type of automation
- Explain the principles of automation

Describe the general histogram characteristics of the automation

UniteFive:Safe work environment

- Apply Work practices of OccupationalHealth Safety (OHS)
- Conduct appropriate techniques to clean Spills
- DiscusWastes generation and segregation
- Explain Safe disposal of bio-hazardous materials and other laboratory wastes

UniteSix:Laboratory records

- Complete entries on report forms and into computer systems all required data.
- Apply Instrument maintenance logs
- Apply all clinical information control and Manage Security and confidentiality

Module Code and Title	<u>HLT MLT4 M04 1224</u> **serological test**
Nominal Duration:	**200** Hours

Module Description: This module covers the knowledge, skills and attitude required to identified concepts of immunology, serology, perform different serological tests and procedures. .

TRAINEE'S OUTCOMES

At the end of the module the learner will be able to:

- Identify concept of immunology and serology
- Apply request paper interpretation and Process samples
- Perform requested tests
- Maintain laboratory records
- Maintain Safe working environment

Module Contents:

MODULE CONTENTS:

Unit-1: Concept of immunology and serology

1.1. Overview of Immune system
1.2. Concepts of antigen and antibody
1.3. Principle of antigen and antibody reaction
1.4. Factors affecting antigen and antibody reaction
1.5. Tests methods in serology

Unit-2: Process serological sample

2.1. Test request review
2.2. Collection ,Preparation and Transportation
2.3. Sample acceptation and rejection criteria
2.4. Processing and storing sample

Unit-3: Tests request

3.1. Selecting tests
  3.1.1. Agglutination test
  3.1.2. Precipitation test
  3.1.3. Complement fixation test
  3.1.4. Immunochromatographic
  3.1.5. ELISA

3.1.6. RIA

3.2. Conducting serological tests.

3.2.1 Common Serologic Tests for Bacterial Infections

3.2.2 Common Serologic Tests for Viral Infections

3.2.3 Serology of Rheumatoid Factor

3.2.4 Serology of C-reactive protein

3.2.5 Serology of human Chorionic Gonadotrophic Hormone (hCG)

3.2.6 Quality assurance in serological analysis

3.3. Making dilutions

3.3.1.Simple dilution

3.3.2. Serial dilution

3.4. Recording, interpreting and verifying

Unit-4: laboratory records

4.1. Entries on report forms or into system

4.2. Maintain instrument logs

4.3. Maintain records of samples received.

4.4. Maintain security and confidentiality

Unit-5: Safe work environment

5.1. Safety work practices

5.2. Waste management

5.3. Handling of specimens and equipment

Learning Methods:

- Interactive lecture
- Group discussion
- Demonstration
- Role playing
- Case study
- Practice with exercise
- Problem based learning

Assessment Methods:

- Continuous assessments (quiz, assignment, tests etc.).
- Oral questioning (Interview)
- Presentation
- Practical demonstration
- Practical assessment by direct observation of tasks using prepared checklist
- Objective based clinical examination
- Final written exam

Assessment Criteria

Unit-1: Identify concept of immunology and serology

- Identify immune system
- Describe concept of antigen and antibody
- State Principle of antigen and antibody reaction
- Identified factors affection antigen and antibody reaction
- Describe methodology of serological tests

Unit-2: Processing serological sample

- Checking Samples and request forms matched before they are accepted.
- Return samples and request forms that do not comply with requirements to their source with reasons for non-acceptance.
- Acceptable samples are logged by applying required document tracking mechanisms.
- Samples are processed as required by requested tests.
- Sample components are stored appropriately until required for testing

Unit-3: Performing requested tests

- Select authorized tests that are indicated for the requested investigations.
- Conduct individual serological tests according to documented methodologies, applying required quality control procedures.
- Perform serial dilution to determine the titer of a sample.
- Record all results, noting any phenomena that may be relevant to the interpretation of results.
- When result interpretation is outside parameters of authorized approval seek discussion with colleague

- Verified results before releasing for clinician/client
- Tested sample or sample components are stored according to organizational sample retention policy for retesting when requested

Unit-4: Maintaining laboratory records

- Making entries on report forms or into computer systems/laboratory information system
- Maintaining instrument logs
- Maintaining records of samples received.
- Maintaining security and confidentiality of client information

Unit-5: Maintain a safe environment

- Using established work practices and PPE
- Minimizing waste generation
- Ensuring the safe disposal of biohazard materials and other laboratory wastes.

Module Code and Title:	HLT MLT4 M05 1224 Clinical Chemistry Tests
NOMINAL DURATION:	250 Hours

MODULE DESCRIPTION: This module covers the knowledge, skills and attitude required to identify concepts of human physiology, anatomy of organs, clinical chemistry, and principle of testing methodology. In addition, interpret clinical chemistry test requirements, prepare samples, conduct pre-use and calibration checks on equipment and perform routine chemical tests/procedures, including data processing and interpretation of results and tracking of obvious test malfunctions where the procedure is standardized.

LEARNING OUTCOMES

At the end of the module the learner will be able to:

- **LO1.** Identify concept of clinical chemistry
- **LO2.** Review test requirements
- **LO3**. Process samples
- **LO4**. Check equipment before use
- **LO5**. Perform Clinical Chemistry Tests
- **LO6**. Process and interpret data
- **LO7.** Maintain laboratory records
- **LO8.** Maintain a safe work environment

MODULE CONTENTS:

LO1. Identify concept of clinical chemistry

1.1. Identifying concept of physiology and anatomy of organs
1.2. Identifying Chemical principles and concepts
1.3. Identifying factors affecting chemical reaction
1.4. Identifying testing methodology

LO2. Review test requirements

2.1. Reviewing test request
2.2. Identifying hazards
2.3. Planning work sequences

LO3. **Process samples**

3.1. Checking samples and request forms
3.2. Returning samples and request forms that do not comply with requirements

3.3. Logging acceptable samples

3.4. Processing samples

3.5. Storing sample components

LO4. Check equipment before use

4.1. Setting equipment/instruments

4.2. Performing pre-use and safety checks

4.3. Identifying faulty or unsafe components

4.4. Checking equipment calibration

4.5. Identifying out of calibration equipment/instruments

4.6. Ensuring availability of reagents.

LO5. Perform Clinical Chemistry Tests

5.1. Selecting authorized tests

5.2. Conducting blood sugar tests

5.3. Conducting liver panel tests

5.4. Conducting renal panel tests

5.5. Conducting lipid panel test

5.6. Recording all results.

5.7. Discussing with colleague for panic results

5.8. Verifying results before releasing

5.9. Storing tested samples or sample components

LO6. Process and interpret data

6.1. Recording test data

6.2. Conducting and computing calibration graphs

6.3. Calculating consistency of values

6.4. Recording results

6.5. Estimating and documenting uncertainty of measurements

6.6. Reporting out of specification or atypical results

6.7. Identifying faulty procedure or equipment problems

LO7. Maintain laboratory records

7.1. Entering approved data into laboratory information system

7.2. Maintaining security and confidentiality

7.3. Maintaining equipment and calibration logs

LO8. Maintain a safe work environment

8.1. Using safety work practices and PPE

8.2. Minimizing generation of wastes and their environmental impacts

8.3. Ensuring safe collection of laboratory and hazardous wastes

8.4. Storing equipment and reagents

LEARNING METHODS:

- Lecture
- Demonstration
- Group discussion
- Exercise
- Individual assignment

ASSESSMENT METHODS:

- Written exam/test
- Practical assessment
- Questioning or interview
- Oral examination

ASSESSMENT CRITERIA:

LO1. Identify concept of clinical chemistry

- Concept of physiology and anatomy of organs is identified
- Chemical principles and concepts are identified
- Factors affection chemical reaction are identified
- Testing methodology of clinical chemistry is identified

LO2. Review test requirements

- Test request are reviewed to identify samples to be tested, test method and equipment/instruments involved
- Hazards and enterprise control measures are identified associated with the sample, preparation/test methods, reagents and/or equipment

- Work sequences are planned to optimize testing of multiple samples

LO3. **Process samples**

- Samples and request forms are checked and matched before they are accepted.
- Samples and request forms that do not comply with requirements are returned to their source with reasons for non- acceptance.
- Acceptable samples are logged, applying required document tracking mechanisms.
- Samples are processed as required by requested tests.
- Sample components are stored appropriately until required for testing

LO4. **Check equipment before use**

- Equipment/Instruments are set up in accordance with test method requirements
- Pre-use and safety checks are performed in accordance with relevant enterprise and operating procedures
- Faulty or unsafe components and equipment are identified and report to appropriate personnel
- Equipment calibration is checked using specified standards and procedures
- Out of calibration equipment/instruments are identified
- Availability of reagents in sufficient quality and quantity is ensured.

LO5. **Perform Clinical Chemistry Tests**

- Authorized tests that are indicated for the requested investigations are selected.
- Blood sugar tests are conducted according to documented methodologies, applying required quality control procedures.
- Liver panel tests are conducted according to documented methodologies, applying required quality control procedures
- Renal panel tests are conducted according to documented methodologies, applying required quality control procedures
- Lipid panel test is conducted according to documented methodologies, applying required quality control procedures
- All results, noting any phenomena that may be relevant to the interpretation of results are recorded.
- When result interpretation is outside parameters of authorized approval is discussed

with colleague

- Results are verified before releasing for clinician/client
- Tested sample or sample components is/are stored according to organizational sample retention policy for retesting when requested

LO6. Process and interpret data

- Test data are recorded by noting atypical observations
- Calibration graphs are constructed, and results for samples computed from these graphs when appropriate.
- Consistency of calculated values is ensured with expectations
- Results are recorded and reported in accordance with enterprise procedures
- Uncertainty of measurement is estimated and documented in accordance with enterprise procedures
- Out of specification or atypical results are reported promptly to appropriate personnel
- If faulty procedure or equipment problems have led to atypical data or results is/are identified

LO7. Maintain laboratory records

- Approved data are entered into laboratory information management system
- Confidentiality and security of enterprise information and laboratory data are maintained.
- Equipment and calibration logs are maintained in accordance with enterprise procedures

LO8. Maintain a safe work environment

- Established safety work practices and PPE are used to ensure personal safety and that of other laboratory personnel OHS
- The generation of wastes and environmental impacts are minimized
- Safe collection of laboratory and hazardous waste is ensured for subsequent disposal (hazard control measures)
- Equipment and reagents are stored with care as required

Module Code And Title:	**HLT MLT4 M06 1224 Immuno-Haematological Tests**
NOMINAL DURATION:	180 Hours

MODULE DESCRIPTION: This module covers knowledge, skills and attitude required to identify concepts of immunohematology, immunology, perform blood typing, cross matching routine and anti-immunoglobulin test and procedures that are part of the requirements of pre- and post-blood transfusion practice.

Training Outcomes

At the end of the module the learner will be able to

- Identify concept of Immunohematology
- Process samples and associated request forms
- Perform tests
- Maintain a safe environment
- Perform Blood Donation and Collection
- Identify Hemolytic Disease
- Maintain laboratory records

MODULE CONTENTS:

Unit one: Basics of Immunohematology

1.1.Introduction to Immunohematology
1.2. blood components and products
1.3. blood genetics and inheritance
1.4.ABO blood group system
1.5. Rh blood group system
1.6.Other blood group system

Unit two: Process immunohematological samples

2.1. Sample acceptance and Rejection criteria
2.2. Sample Processing

Unit three: Types of Immuno Haematological tests

3.1. ABO grouping/typing
3.2. RH typing

3.3. Compatibility test/Cross matching

3.4. Anti-immunoglobulin test

Unit four Blood Donation and Collection

14.1 selection of blood donors

14.2 collection of blood and preservatives

14.3 Blood component processing

Unit Five :- safe work environment

5.1 Establish OHS work practices and PPE

5.2 Waste management and safe disposal

Unit Six :- Hemolytic Disease

6.1.Hemolytic trasnfusion reaction

6.2.Autoimmune Hemolytic Anemia (AIHA)

6.3. Hemolytic Disease of the Fetus and New born (HDFN

6.4 Rho (D) Immune Globulin (Human) – RhIG

6.5.Treatment of Infants Suffering from HDFN

Unit seven:- Laboratory Record Management

7.1.Entering approved data in to computer systems

7.2.instrument logs Maintenance

7.3.records of blood and blood products

7.4.Maintaining security and confidentiality

LEARNING METHODS:

Learning Methods:

- Interactive lecture
- Group discussion
- Demonstration
- Group assignment
- Individual assignment

ASSESSMENT METHODS:

- Continuous assessments (quiz, assignment, tests etc.).

- Oral questioning (Interview)
- Skill check out
- Final written exam

Assessment Criteria

Unit one: Basics of Immunohematology

- Identify concept of Immunohematology
- Describe blood components and products
- Describe blood genetics and inheritance
- Describe ABO blood group system
- Describe Rh blood group system
- Describe Other blood group system

Unit two: Process immunohematological samples

- Identify Sample acceptance and Rejection criteria
- Perform Sample Processing

Unit three: Types of Immuno Haematological tests

- Perform ABO grouping/typing
- Perform RH typing
- Perform Compatibility test/Cross matching
- Perform Anti-immunoglobulin test

Unit four Blood Donation and Collection

- Identify selection of blood donors
- Perform collection of blood and preservatives
- Describe Blood component processing

Unit Five :- Maintain safe work environment

- Establish OHS work practices and PPE
- Apply Waste management and safe disposal

Unit Six :- Unit Six :-Describer Hemolytic Disease

- Identify Hemolytic trasnfusion reaction
- Identify Autoimmune Hemolytic Anemia (AIHA)
- Describe Hemolytic Disease of the Fetus and New born (HDFN
- Identify Rho (D) Immune Globulin (Human) – RhIG

- Describe Treatment of Infants Suffering from HDFN

Unit seven :- Laboratory Record Management

- Entering approved data in to computer systems
- Perform instrument logs Maintenance
- Perform records of blood and blood products
- Maintaining security and confidentiality

Module Code and Title	**HLT MLT4 M07 1224 Histopathological Samples for Examination**
Nominal Duration:	140 Hours

Module Description: This module covers the knowledge, skills and attitude required to prepare histological and pathological samples for examination involving processing and sectioning of human tissues.

TRAINEE'S OUTCOMES

At the end of the module the learner will be able to:

- Describe about histology
- Assemble equipment and materials
- Process tissue
- Stain tissue sections
- Maintain a safe work environment

Module Contents:

MODULE CONTENTS:

UNIT-1: Introduction to histology

1.1. Diagnostic techniques used in pathology

1.2.Causes of Diseases

Unit -Two Equipment and materials setup

1.6. Number and type of required sections

1.7. Arrangement of workspace equipment

1.8. Performing pre-use and safety checks

1.9. Reporting faulty or unsafe equipment

1.10. Inspection of processor reagents

1.11. Assembling of equipment's, safety materials and containers

Unit-2: Tissue Processing

2.5. Preparation of fine structure tissue

2.6. Performing tissue section

2.7. Mounting tissue sections on microscopic slide

2.8. Quality of embedded tissue

Unit-3: Stain sections

3.2. Selecting reagents

3.3. Staining tissue sections

3.4. Examining sections microscopically

3.5. Preparing mounted section permanently

3.6. Photograph and present Section

3.7. Attachment of permanent labels

3.8. Security and confidentiality of client information

Unit-4: Safe work environment

4.5. Personal safety

4.6. Handling of specimens and equipment

4.7. waste management

4.8. Hazards and incidents

Learning Methods:

- Interactive lecture
- Group discussion
- Demonstration
- Role playing
- Case study
- Practice with exercise
- Problem based learning

Assessment Methods:

- Continuous assessments (quiz, assignment, tests etc.).
- Oral questioning (Interview)
- Presentation
- Practical demonstration
- Practical assessment by direct observation of tasks using prepared checklist
- Objective based clinical examination
- Final written exam

Assessment Criteria

Unit One: Introduction to histology
- Identify the causes of Disease
- Identify diagnostic techniques used in pathology

Unit Two: Assemble equipment and materials
- Confirm number and type of required sections
- Collect and arrange workspace equipment
- Perform pre-use and safety checks
- Report faulty or unsafe equipment
- Inspect processor reagents and report any items require replacement
- Assemble specific process equipment's, safety materials and containers

Unit-2: Process tissue
- Prepared fine Structure of tissue
- Perform tissue sectioning
- Mounte tissue section on microscopic slide
- Check the quality of embedded tissue is

Unit-3: Stain sections
- Selection of reagents
- tissue sections Staining
- Examine tissue sections microscopically
- Prepare mounte section permanently
- Identify photograph and present Section
- Perform attachment of permanent labels
- Identify security and confidentiality of client information

Unit-4: Safe work environment

- Ensure Personal safety and cross-contamination minimizing
- Identify specimens and equipment Handling
- Generation of waste and environmental impacts is minimized
- Reporting of hazards and incidents

Module Code and Title	HLT MLT4 M08 1224 Quality Assurance
Nominal Duration:	**90** Hours

- **MODULE DESCRIPTION:** This module covers the knowledge, skills and attitude required to identify the Concept quality assurance, prepare document and record, implement quality Pre-analytic process, implement quality analytic process, implement quality post- analytic process and conduct Process improvement activities.

TRAINEE'S OUTCOMES

At the end of the module the learner will be able to:

- Identify the Concept quality assurance
- Prepare document and record
- Implement quality Pre-analytic process
- Implement quality analytic process
- Implement quality post- analytic process
- Conduct Process improvement activities

Module Contents:

MODULE CONTENTS:

Unit 1: Concept quality assurance

1.1. Identifying concepts of quality assurance
1.2. Identifying differences of quality assurance and quality control
1.3. Benefit of quality assurance program
1.4. Identifying quality elements in all quality cycle
1.5. Identifying laboratory path of work flow

Unit 2: Preparation of document and record

2.1. Identifying document and record systems
2.2. Preparing SOP and guidelines
2.3. Accomplishing quality manual

Unit 3: Implementation quality Pre-analytic process

3.1 Identifying patient and laboratory requests

3.2 labeling specimen and requests

3.3 Collecting the specimen

3.4 Identifying and maintaining chain of custody

3.5 Receiving, storing and transporting specimen

Unit 4: Implementation quality analytic process

4.1. Identifying internal quality controls and calibrated materials

4.2. Differentiating error, accuracy and precision

4.3. Defining qualitative and quantitative quality controls

4.4. Performing Internal and external quality control

4.5. Calculating and interpreting quality control data

4.6. defining biological reference range and critical values

unit 5: Implementation quality post- analytic process

5.1 Interpreting test result

5.2 Identifying reporting system

5.3 Releasing of test result

5.4 Maintaining client's information and confidentiality

Unit 6:**implementation of process improvement activities**

6.1 Identifying laboratory related occurrences

6.2 Assessing root cause of the occurrence

6.3 Taking corrective and prevented action

Learning Methods:

- Interactive lecture
- Group discussion
- Role playing
- Case study
- Practice with exercise
- Problem based learning

Assessment Methods:

• Continuous assessments (quiz, assignment, tests etc.). • Oral questioning (Interview) • Presentation • Practical assessment by direct observation of tasks using prepared checklist • Final written exam
Assessment Criteria

Unit one: Identify the Concept quality assurance

- Concepts of quality assurance is identified
- Differences of quality assurance and quality control are identified
- Benefit of quality assurance program is understood
- Identify quality elements in pre-analytical, analytical and post analytical laboratory process.
- Laboratory path of work flow is identified

Unit two: Prepare document and record

- Document and record systems are identified.
- SOP and guidelines are prepared
- Records are identified, achieved and indexed according to document policy of the organization
- Accomplish all laboratory activities according to quality manual

Unit three: Implement quality Pre-analytic process

- Patient and laboratory requests are properly identified.
- Specimen and requests are properly labeled according to the laboratory procedures
- Collect the right specimen at the right time with proper collection materials according to the laboratory procedures
- Chain of custody are identified and maintained
- Proper specimen are received, stored and transported according to quality policy manuals.

Unit four: Implement quality analytic process

- Identify Internal quality controls and calibrated materials
- Differentiate Error, accuracy and precision
- Define Qualitative and quantitative quality controls
- Identify Qualitative quality control methods
- Perform Internal quality control
- Calculate and interpret quality control data
- Identify External quality control methods

Unit five: Implement quality post- analytic process

- interpret test results according to the laboratory procedures
- Identify proper reporting system according to the laboratory procedures
- Release test results according to the laboratory procedures
- maintain information and confidentiality of client test results according to the laboratory information management policy

LO6. Conduct Process improvement activities

- Laboratory related occurrences are identified
- Root cause of the occurrence are assessed

Unit six: implement Process improvement activities

- Identify Laboratory related occurrences
- Assess Root cause of the occurrence
- Take corrective and prevented actions using different tools.
- Identify continual improvement methods

Module code and title :	HLT MLT4 M09 1224 Managing Community Health Service
Nominal Duration :	100 Hours:

MODULE DESCRIPTION: This module describes the knowledge, skills and attitude required to manage health service of the area to improuve quality of Heath service.

LEARNING OUTCOMES

At the end of the module the learner will be able to:

- Describe organizational guidelines, understand health policy and service delivery system
- Understand Plan, manage, monitor and evaluate health system
- Explain Lead and build individual's and team's capacity

MODULE CONTENTS:

Unit One: Organizational guidelines, understand health policy and service delivery system

1.1. Policy and organization of the health care system of Ethiopia
1.2. Primary healthcare in Ethiopia
1.3. Elements of primary health care
1.4. Health service extension program
1.5. Following workplace instructions and policies
1.6. organizational programs and procedures
1.7. Utilizing organizational resources

Unit Two: Plan, manage, monitor and evaluate health system

2.1 Management skills and efficient health care system.
2.2 Planning health programs.
2.3 Managing healthcare resources.
2.4 Principles of health care ethics
2.5 Developing Individual and team capacity
2.6 Resolving issues through participation and consultation
2.7 Health service monitoring and evaluation mechanisms

Unit Three: Lead and build individual's and team's capacity

3.1 Identifying self-improvement areas based on individual's self-performance evaluation.
3.2 Identifying and implementing learning and development needs

3.3 learning and developing program goals and objectives

3.4 Providing workplace learning opportunities and coaching/ mentoring

3.5 Developing team and individuals for joint action plans.

3.6 Allocating duties and responsibilities

3.7 Making collaborative efforts to attain organizational goals

LEARNING METHODS:

- Lecture and discussion
- Role play
- Group discussions
- Individual assignments
- Demonstration
- Field visit

ASSESSMENT METHODS:

- Written exam/test
- Questioning
- Demonstration and observation
- Direct observation of tasks(real or simulated practice) using checklist
- Review of tasks (portfolio, logbook, report, assignment…)

ASSESSMENT CRITERIA

Unit One: Organizational guidelines, understand health policy and service delivery system

- Identify the policy and organization of the health care system of Ethiopia
- Explore primary healthcare in Ethiopia
- Identify elements of primary health care
- Present health service extension program
- Follow workplace instructions and policies
- Identify organizational programs and procedures
- Utilize organizational resources

Unit Two: Plan, manage, monitor and evaluate health system

- Identify management skills and efficient health care system.
- Plan health programs.
- Manage healthcare resources.
- Identify principles of health care ethics
- Develop Individual and team capacity
- Resolve issues through participation and consultation
- Develop health service monitoring and evaluation mechanisms

Unit Three: Lead and build individual's and team's capacity

- Identify self-improvement areas based on individual's self-performance evaluation.
- Identify and implement learning and development needs
- Learn and develop program goals and objectives

Module Title	HLT MLT4 M09 1224 **Preventing and Eliminating MUDA**
Module Code	**HLT MLT4 M10 1224**
Nominal Duration :	**56 Hours**

Module Description : This module covers the knowledge, skills and attitude required by a worker to prevent and eliminate MUDA/wastes in his/her work place by applying scientific problem-solving techniques and tools to enhance quality, productivity and other kaizen elements on continual bas is. It covers responsibility for the day-to-day operation of the work and ensures Kaizen Elements are continuously improved and institutionalized.

Training Outcomes

At the end of the module the trainee will be able to attain the following training objectives:

- Identification of requirements
- Identification of MUDA
- Causes of problem.
- MUDA elimination
- MUDA prevention

Module Contents:

Unit one: Identification of requirements

1.1.Work instructions
1.2.Job specifications
1.3.OHS
1.4.Selecting of materials
1.5.Safety tools and equipment

Unit two: Identification of MUDA

2.1. Planning
2.2. Causes and effects of MUDA
2.3. Tools and techniques
2.4. Kaizen Board
2.5. Procedures

2.6. Reporting

Unit three: Causes of problem.

3.1.Listing causes

3.2.4M1E

3.3.Idea generation

3.4.Solutions test and evaluation

3.5.Action plan

Unit four: MUDA elimination

4.1.Planning

4.2.Action plan

4.3.Principles for improvement

4.4.Tools and techniques

4.5.Tangible and intangible results

4.6.Types of diagrams

4.7.Improvements

4.8.reporting

Unit five: MUDA prevention

5.1.Planning

5.2.Standardizing

5.3.Methods of prevention

5.4.Standard Operating Procedures (SOPs)

5.5.Problem selection

Learning Methods:

- Lecture
- Group discussion
- Demonstration
- Brainstorming

Assessment Methods:

- Written test
- Oral questioning
- Practical demonstration
- Observation in prepared checklist

Assessment Criteria:
Unit one: Identification of requirements • Use work instructions • Read and interpret job specifications • Identify and check safety tools and equipment **Unit two: Identification of MUDA** • Prepare and implement planning • Discuss causes and effects of MUDA • Identify Kaizen elements • Identify and measure procedures **Unit three: Causes of problem.** • List causes • Analyze 4M1E • Suggest solutions test and evaluation **Unit four: MUDA elimination** • Adopt principles for improvement • Identify tangible and intangible results • Compare types of diagrams • Report improvements **Unit five: MUDA prevention** • Prepare and implement planning • Discuss and prepare standardizing • Understand methods of prevention • Ensure Standard Operating Procedures (SOPs)

3. Resources

Item No.	Category/Item	Description/ Specifications	Quantity	Recommended Ratio (Item: Learner)
A.	**Learning Materials**			
1.	TTLM	• Prepared by the trainer	25	1:1
2.	Textbooks	-	5	1:5
2.1	Immunohematology: Principles and Practice	Eva D. Quinley , 3rd edition: 2011		
3.	Reference Books			
3.1.	Immunohematology for Medical Laboratory Technicians	Sheryl A. Whitlock; 2010	5	1:5
3.2.	Immunohematology and Transfusion Medicine: A Case Study Approach	Mark T. Friedman, Kamille A. West Peyman Bizargity, Kyle Annen Jeffrey S. Jhang. Second Edition; 2015		
3.4.	Rossi's Principles of Transfusion Medicine	Wiley Blackwell .5th edition; 2016		
4.	**Learning Facilities & Infrastructure**	• Health Indicators/latest • EDHS,2016 • Fact sheets • Standard formats	10	1:3
B.	**Learning Facilities & Infrastructure**			
1.	Lecture Room	5*5m	1	1:25

2.	Library	Standard	1	
3.	Demonstration room	5*10m	1	1:25
C.	**Consumable Materials**			
1.	Paper	A4	5rim	1:5
2.	Pen	Standard		
3	Pencil and rubber	Standard		
4	Graph paper	Standard		
5	Bucher paper	Standard	10	1:3
6	Marker	Standard	12 per pack	
7	Printer ink	Standard	4	
8	White board marker	Standard	15	
9	Plaster	Standard	1 roll	
10	Gloves	Standard	1 pack	
11	70% alcohol	Standard	1 bottle	
12	Disinfectant solution	Standard	1 bottle	
13	Anti-sera A, B and D	Standard	1 vial each	
14	Anti-human globulin	Standard	1 vial	
15	Cotton	Standard	1 roll	

16	Gauze	Standard	1 roll	
17	Physiological saline	Standard	1 bottle	
18	Sealing clay	Standard	1 pc	
19	Capillary tubes	Standard	1 pak	
20	Blood lancet	Standard	1 pak	
21	Microscopic slides	Standard	1pak of 50	2:1
22	Test tubes	Standard	25	1:1
23	Anticoagulants	Standard	1 bottle	
24	Needle with syringes	Standard	1 pak	
D.	**Tools and Equipment**			
1.	Computer	Lap top	1	1:25
2.	LCD projector	LCD Projector	1	1:25
3.	Printer	Laser Jet	1	1:25
4	Photocopy machine	Standard	1	1:25
5	Scanner	Standard	1	1:25
6	Back up	Standard	1	1:25
7	White board	110X80mm	1	1:25
8	Microscope	Standard	10	1:3
9	Macro centrifuge	Standard	2	1:25
10	Macro centrifuge	Standard	1	1:25
11	Water bath	Standard	1	1:25

12	ELISA machine	Standard	1	1:25
13	Incubator	Standard	1	1:25
14	Shaker	Standard	1	1:25
15	Refrigerator	Standard	1	1:25
16	Plates (ELISA)	Standard	1	1:25
17	Glass wares	Standard	1	1:25
18	Auto clave	Standard	1	1:25
19	Weighing Scale	Standard	1	1:25
20	Thermometer	Standard	1	1:25
21	Test tubes	Standard	1	1:25
22	Tourniquet	Standard	1	1:25
23	Container for sharp instruments	Standard	1	1:25

Developers Profile

No	Name	Qualification (Level)	Field of Study	Organization/ Institution	Mobile number	E-mail
1	Furo Beshir Nebi	MSc	Tropical and Infectious Disease	Harar Health Science College	0911173970	nebi.furo@gmail.com
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6	Melaku Gudeta	1st Degree in MLT	MLT	Mettu HSC	0912181999	melakugudeta@gmail.Com
7						